

Device Authentication Through Deep Learning-Based RF Fingerprinting

ECE/CS 478 Network Security

Final Project Report

Jorge Gonzalez Lopez

Group 10

Oregon State University

Spring 2026

Abstract—This project investigates device authentication through deep learning-based RF fingerprinting. A Convolutional Neural Network (CNN) was trained to identify 31 Bluetooth Low Energy (BLE) devices using RF signal samples. The model was evaluated under same-domain and cross-domain conditions across different RF channels and physical locations. Raw IQ samples were first used as input features. Then, phase-derivative features were extracted from the IQ samples to evaluate whether they improved robustness against domain shift. The results showed nearly perfect accuracy under same-domain conditions, proving that the CNN successfully learned device-specific RF fingerprints when training and testing conditions matched. However, raw IQ samples performed poorly across RF channels, showing strong sensitivity to channel variation. Cross-location testing performed better than cross-channel testing, indicating that physical location changes were less damaging than channel changes. Phase-derivative features greatly improved cross-domain performance, especially across channels. Overall, the project demonstrates that RF fingerprinting is a promising method for wireless device authentication, but real-world systems must address domain shift across channels and environments.

I. INTRODUCTION

Radio Frequency (RF) fingerprinting is a physical-layer device authentication technique that uses unique hardware imperfections in wireless transmitters. Due to manufacturing differences, no two radio devices are exactly identical. Small variations in oscillators, power amplifiers, mixers, and filters create subtle distortions in transmitted RF signals. These distortions can act like fingerprints that help identify a specific device.

Traditional authentication methods such as passwords, MAC addresses, and cryptographic credentials can be stolen, spoofed, or copied. RF fingerprinting provides an additional layer of security because it relies on physical hardware characteristics that are difficult to duplicate. As wireless systems continue to expand through Bluetooth Low Energy (BLE), Internet of Things (IoT) devices, and cyber-physical systems, secure device authentication has become increasingly important.

Deep learning-based RF fingerprinting combines machine learning with RF signal analysis to identify wireless devices based on their unique signal characteristics. Each wireless device produces slightly different RF signals due to hardware imperfections, and deep learning models can learn these subtle patterns to distinguish one device from another. Convolutional Neural Networks (CNNs) are particularly effective because they automatically extract discriminative features from raw RF signal samples without extensive manual feature engineering.

In a typical RF fingerprinting system, RF signals are collected from wireless devices, processed into a suitable representation, and then provided to a CNN model for training and classification. The model learns device-specific characteristics and can determine whether a received signal originated from a trusted or unknown device. This capability makes RF fingerprinting a promising technology for improving the security of wireless communication systems.

The goal of this project is to evaluate whether a CNN can identify 31 Bluetooth Low Energy devices using RF fingerprints. The project also studies how performance changes when training and testing occur across different channels

and locations. Finally, phase-derivative features are tested to determine whether they improve stability under changing conditions.

II. BACKGROUND

RF fingerprinting works because every transmitter introduces small device-specific distortions into its emitted signal. These distortions come from hardware components such as oscillators, power amplifiers, mixers, and filters. Even if two radios are designed to transmit the same signal, manufacturing tolerances cause each device to slightly alter the transmitted waveform in a unique way.

The RF signals in this project are represented using IQ samples. The I component is the in-phase part of the signal, while the Q component is the quadrature part. Together, I and Q describe the received signal waveform and contain information about the transmitting device. Since the I and Q components preserve both amplitude and phase information, they are useful for machine learning-based device classification.

CNNs are useful for RF fingerprinting because they can automatically learn patterns from raw signal samples. Instead of manually designing features, the CNN extracts useful characteristics through convolutional layers and uses them to classify the device.

A major challenge is domain shift. A model trained on one channel or location may perform worse when tested on another channel or location because the signal changes due to channel effects, propagation, interference, or environmental differences.

III. DATASET DESCRIPTION

The dataset contains Bluetooth Low Energy transmissions from 31 devices. Data was collected across four RF channels and four physical locations. The channel domains were Ch1_R1, Ch2_R1, Ch14_R1, and Ch32_R1. The location domains were Loc1, Loc2, Loc3, and Loc4.

TABLE I
DATASET DOMAINS

Category	Domains
Channels	Ch1_R1, Ch2_R1, Ch14_R1, Ch32_R1
Locations	Loc1, Loc2, Loc3, Loc4
Devices	31 BLE Devices

Each folder contained training and testing files: X_train.npy, Y_train.npy, X_test.npy, and Y_test.npy. The X files contained IQ signal samples, and the Y files contained the device labels. Each original RF frame contained IQ samples from the received BLE signal. For the phase-derivative experiments, the first 128 IQ samples were used, producing 127 phase-derivative values after differentiation.

IV. CNN METHODOLOGY

The CNN used RF signal samples as input. The model included one-dimensional convolutional layers, max-pooling layers, a flatten layer, dense layers, and a final softmax output layer.

TABLE II
CNN ARCHITECTURE

Layer	Configuration
Input	IQ samples or phase-derivative features
Conv1D	32 filters, kernel size 3, ReLU
MaxPooling1D	Pool size 2
Conv1D	64 filters, kernel size 3, ReLU
MaxPooling1D	Pool size 2
Flatten	Converts features to one vector
Dense	128 neurons, ReLU
Output	31 neurons, Softmax

TABLE III
TRAINING PARAMETERS

Parameter	Value
Model Type	1D CNN
Classes	31
Optimizer	Adam
Loss Function	Categorical Cross-Entropy
Batch Size	64
Epochs	5

V. TASK 1: SAME-DOMAIN RESULTS

Task 1 evaluated the baseline performance of the CNN under same-domain conditions. For each experiment, the model was trained and tested using data collected from the same RF channel or physical location. The objective was to determine the maximum achievable classification performance when no domain shift was present.

A. Experimental Setup

Eight same-domain experiments were completed. Four experiments used channel-based domains, and four experiments used location-based domains. The channel experiments were Ch1_R1 to Ch1_R1, Ch2_R1 to Ch2_R1, Ch14_R1 to Ch14_R1, and Ch32_R1 to Ch32_R1. The location experiments were Loc1 to Loc1, Loc2 to Loc2, Loc3 to Loc3, and Loc4 to Loc4.

TABLE IV
SAME-DOMAIN ACCURACY RESULTS

Domain	Accuracy (%)
Ch1_R1	100.00
Ch2_R1	100.00
Ch14_R1	99.99
Ch32_R1	100.00
Loc1	100.00
Loc2	100.00
Loc3	99.98
Loc4	100.00

B. Results and Analysis

The CNN achieved nearly perfect classification accuracy across all eight same-domain experiments. Channel-based accuracies ranged from 99.99% to 100%, while location-based accuracies ranged from 99.98% to 100%. The highest accuracies were observed for Ch1_R1, Ch2_R1, Ch32_R1, Loc1,

Loc2, and Loc4, all of which achieved perfect classification performance.

The slightly lower accuracies observed for Ch14_R1 and Loc3 were negligible and indicate only a very small number of misclassifications. Overall, the model demonstrated strong capability in learning the distinguishing characteristics of each BLE transmitter.

C. Importance of Same-Domain Testing

Same-domain testing serves as the baseline for evaluating RF fingerprinting performance. Because the training and testing data originate from identical environments, the model is exposed to consistent signal characteristics during both phases. This removes external sources of variation and allows the CNN to focus on learning device-specific RF fingerprints.

The results obtained in Task 1 establish an upper performance bound against which cross-domain experiments can later be compared. Since the same-domain results were nearly perfect, any performance decrease in later tasks can be attributed to domain shift rather than the CNN being unable to learn the devices.

D. Task 1 Summary

Task 1 established the baseline performance for the RF fingerprinting system. The CNN successfully identified all 31 BLE devices with accuracies approaching 100% across every same-domain experiment. These findings demonstrate that RF fingerprints can be highly effective for device identification when environmental and channel conditions remain unchanged. In this task, we prove that under ideal conditions, the CNN can achieve nearly perfect device classification accuracy and successfully learn unique, device-specific RF fingerprints when training and testing conditions match.

VI. TASK 2: CROSS-DOMAIN RAW IQ RESULTS

Task 2 evaluated the ability of the CNN to generalize across different domains using raw IQ samples. Unlike Task 1, the model was trained on one RF channel or location and tested on a different channel or location. This introduced domain shift, allowing the robustness of the learned RF fingerprints to be evaluated under changing conditions.

A. Experimental Setup

Two categories of experiments were performed. The first category examined cross-channel performance, where training and testing were conducted on different BLE channels. The second category examined cross-location performance, where training and testing were conducted at different physical collection locations. A total of 24 cross-domain raw IQ experiments were completed.

B. Cross-Channel Raw IQ Results

The cross-channel experiments produced the lowest accuracies observed in the project. Most channel-to-channel classification accuracies ranged between approximately 3% and 8%. These results indicate that raw IQ features were highly sensitive to channel variations.

TABLE V
CROSS-CHANNEL RAW IQ ACCURACY

Experiment	Accuracy (%)
Ch1 → Ch2	1.92
Ch1 → Ch14	7.89
Ch1 → Ch32	3.83
Ch2 → Ch1	3.39
Ch2 → Ch14	3.91
Ch2 → Ch32	7.45
Ch14 → Ch1	7.67
Ch14 → Ch2	5.00
Ch14 → Ch32	5.10
Ch32 → Ch1	6.80
Ch32 → Ch2	4.37
Ch32 → Ch14	4.83

The poor cross-channel performance suggests that channel characteristics significantly altered the appearance of the received RF signals. As a result, the CNN struggled to recognize device fingerprints when training and testing occurred on different channels.

C. Cross-Location Raw IQ Results

Cross-location performance was considerably stronger than cross-channel performance. Most cross-location experiments achieved accuracies between approximately 66% and 93%. This indicates that device-specific RF characteristics remained partially preserved when the collection location changed.

TABLE VI
CROSS-LOCATION RAW IQ ACCURACY

Experiment	Accuracy (%)
Loc1 → Loc2	93.38
Loc1 → Loc3	70.79
Loc1 → Loc4	75.49
Loc2 → Loc1	88.36
Loc2 → Loc3	69.81
Loc2 → Loc4	71.47
Loc3 → Loc1	66.33
Loc3 → Loc2	73.09
Loc3 → Loc4	83.81
Loc4 → Loc1	74.93
Loc4 → Loc2	73.02
Loc4 → Loc3	80.17

D. Results and Analysis

A major observation from Task 2 is that channel changes had a substantially larger impact on performance than location changes. While location-based experiments frequently maintained moderate to high classification accuracy, channel-based experiments often approached random guessing performance.

These results suggest that the CNN learned features that were strongly influenced by channel conditions. When the channel changed, the distribution of the received signals changed significantly, causing severe degradation in classification performance.

E. Task 2 Summary

Task 2 showed that raw IQ samples are highly vulnerable to domain shift. Cross-channel accuracy was extremely low, while cross-location accuracy was much stronger. This means that changing RF channels damaged model performance more than changing physical locations. These findings motivated the phase-derivative feature extraction approach tested in Task 3.

VII. TASK 3: PHASE-DERIVATIVE RESULTS

Task 3 evaluated whether phase-derivative features could improve cross-domain robustness. Instead of using raw IQ samples directly, the phase of the signal was extracted from the I and Q components. The phase was then unwrapped, and the phase derivative was calculated.

A. Feature Extraction Method

The signal phase was computed as

$$\phi[n] = \tan^{-1} \left(\frac{Q[n]}{I[n]} \right) \quad (1)$$

where $I[n]$ is the in-phase component and $Q[n]$ is the quadrature component. After the phase was calculated, phase unwrapping was applied to remove discontinuities. The phase derivative was then calculated from the unwrapped phase sequence.

This feature emphasizes how the phase changes over time. The goal was to preserve device-specific signal behavior while reducing the effect of channel-dependent distortions.

B. Cross-Channel Phase-Derivative Results

Phase-derivative features greatly improved cross-channel performance compared to raw IQ samples. Several experiments increased from below 10% accuracy to above 85%.

TABLE VII
CROSS-CHANNEL PHASE-DERIVATIVE ACCURACY

Experiment	Accuracy (%)
Ch1 → Ch2	93.16
Ch1 → Ch14	91.70
Ch1 → Ch32	54.76
Ch2 → Ch1	92.30
Ch2 → Ch14	86.87
Ch2 → Ch32	57.58
Ch14 → Ch1	89.48
Ch14 → Ch2	86.46
Ch14 → Ch32	61.42
Ch32 → Ch1	50.06
Ch32 → Ch2	50.84
Ch32 → Ch14	61.21

The strongest improvements occurred when training and testing involved Ch1, Ch2, and Ch14. Performance involving Ch32 was lower than the other phase-derivative results, but it was still much better than the raw IQ cross-channel results.

C. Cross-Location Phase-Derivative Results

Phase-derivative features also improved many cross-location experiments. Most location-based phase-derivative accuracies were above 84%, and several exceeded 93%.

TABLE VIII
CROSS-LOCATION PHASE-DERIVATIVE ACCURACY

Experiment	Accuracy (%)
Loc1 → Loc2	93.68
Loc1 → Loc3	72.00
Loc1 → Loc4	96.30
Loc2 → Loc1	95.06
Loc2 → Loc3	90.00
Loc2 → Loc4	92.29
Loc3 → Loc1	84.16
Loc3 → Loc2	93.46
Loc3 → Loc4	91.18
Loc4 → Loc1	94.91
Loc4 → Loc2	92.27
Loc4 → Loc3	86.42

D. Task 3 Summary

Task 3 showed that phase-derivative features significantly improved robustness under cross-domain conditions. Cross-channel accuracy improved the most. For example, Ch1 to Ch2 increased from 1.92% using raw IQ to 93.16% using phase-derivative features. This demonstrates that phase-derivative feature extraction reduced the negative impact of channel domain shift.

VIII. RAW IQ VS PHASE-DERIVATIVE COMPARISON

The most important comparison in this project is the difference between raw IQ and phase-derivative performance. Raw IQ samples performed extremely well under same-domain conditions, but they failed badly across RF channels. Phase-derivative features produced much stronger cross-domain results.

TABLE IX
RAW IQ VS PHASE-DERIVATIVE EXAMPLE COMPARISON

Experiment	Raw IQ (%)	Phase Derivative (%)	Change
Ch1 → Ch2	1.92	93.16	+91.24
Ch1 → Ch14	7.89	91.70	+83.81
Ch2 → Ch1	3.39	92.30	+88.91
Ch14 → Ch2	5.00	86.46	+81.46
Loc2 → Loc3	69.81	90.00	+20.19
Loc3 → Loc2	73.09	93.46	+20.37

Raw IQ features were strongly affected by channel conditions. This indicates that the CNN may have learned channel-dependent characteristics in addition to device-specific characteristics. When the test channel changed, these learned patterns no longer matched, causing accuracy to drop.

Phase-derivative features improved performance because they focused on changes in signal phase rather than directly using raw IQ values. This helped reduce the effect of channel-dependent distortions and allowed the CNN to better recognize device-specific behavior.

IX. DISCUSSION

The results show that CNN-based RF fingerprinting can accurately identify BLE devices under controlled same-domain conditions. The nearly perfect Task 1 results demonstrate that the dataset contained useful device-specific RF fingerprints and that the CNN architecture was capable of learning them.

However, the Task 2 results show that raw IQ models do not generalize well across RF channels. This is a major limitation for real-world deployment because wireless devices may operate on different channels. If the authentication system is trained on one channel but tested on another, raw IQ features may not be reliable.

Location changes were less damaging than channel changes. This suggests that the CNN learned some device-specific features that remained useful across physical environments. Still, cross-location performance was lower than same-domain performance, which means environmental changes still introduced domain shift.

Phase-derivative feature extraction was the strongest improvement tested in this project. It greatly improved cross-channel accuracy and often improved cross-location accuracy. This shows that feature engineering can help reduce domain shift in RF fingerprinting systems.

X. LIMITATIONS AND FUTURE WORK

This project was limited to 31 BLE devices, four RF channels, and four physical locations. Although the results are useful, a larger dataset would be needed to evaluate how

well the method scales to more devices and more realistic environments.

Another limitation is that the CNN was trained for only five epochs. Additional training, hyperparameter tuning, and deeper model architectures could potentially improve performance. Future work could also include transfer learning, domain adaptation, and multi-domain training to improve robustness across unseen channels and locations.

Confusion matrices would provide additional insight into which devices were most commonly misclassified. Future work should include per-device confusion analysis to identify which transmitters are most robust and which are most vulnerable under domain shift.

XI. CONCLUSION

This project demonstrated that CNNs can identify BLE devices using RF fingerprinting. Same-domain testing achieved nearly perfect accuracy, showing that the model can learn unique device fingerprints. However, raw IQ samples performed poorly across different channels because of strong domain shift.

Phase-derivative features significantly improved cross-domain performance and made the model more robust. The strongest improvement was observed in cross-channel testing, where raw IQ accuracy was often below 10%, but phase-derivative accuracy was frequently above 85%. Overall, RF fingerprinting is a promising technique for wireless device authentication, but real-world systems must address domain shift across channels and environments.

XII. INDIVIDUAL CONTRIBUTIONS

Jorge Gonzalez Lopez contributed to dataset organization, CNN training, same-domain experiments, cross-domain raw IQ experiments, phase-derivative testing, result collection, result analysis, figure preparation, table creation, and report writing.